

FOS Geospatial Tools

A handbook for geospatial practitioners and scientists

Welcome to Free and Open Source Geospatial Tools v 2.0. This book provides resources and examples of open-source geospatial software and solutions and is updated from the version published in 2024.

1. SUPPORT

This book is free to use for anyone. If you want to support, donate, or buy me a coffee, please visit my [paypal](#) site. I will be adding chapters and resources in the future. Please share the book with others who might benefit from reading it. I appreciate your support!

2. WHY THIS BOOK?

Accessing, processing, and analyzing geospatial data can be daunting, often hindered by data accessibility, compatibility, and cost constraints. There are many geospatial resources, tutorials, and books available, but most focus on specific tools, such as Spatial SQL (Forrest, 2023), the Cloud-Based Remote Sensing with Google Earth Engine remote sensing (Cardille et al., 2023) or Qiusheng Wu's excellent geospatial data analysis book using Geemap and Earth Engine (Wu, 2023). Inspired by the Free and Open Source Software for Geospatial (FOSS4G) conference started by the Open Source Geospatial Foundation, Free and Open Source Geospatial Tools provides a guide to open-source geospatial software packages and workflows.

At the software level, many tutorials, GitHub repositories, and software sites assume you are familiar with installation or running code. They'll use acronyms such as CLI and IDE; you will have no idea what they're talking about if you're a beginner. Once you decipher the terms, running into errors and trying to fix them can be trying or a deterrent to using free and open-space geospatial software. Don't be discouraged by entering this new world; keep at it when errors occur, or the code doesn't do what you want.

The FOSS book aims to address these challenges head-on by providing practical solutions, resources for further learning, and tips for getting the basics running smoothly. You want to analyze and visualize data, not spend time looking through Stack Exchange or asking ChatGPT how to fix an obscure error when your code doesn't run, or you can't install a package correctly.

2.a. Audience & Learning

This book is intended for beginners with some knowledge of desktop tools such as ArcGIS Pro or QGIS and a limited understanding of coding using Javascript or Python. Ideally, you will have taken an introductory geography and computer science course that introduced you to software and languages. If not, you can still get started, but some background knowledge, such as basic Python coding, will be needed to get

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I would like to thank and acknowledge many of the free and open-source geospatial pioneers who offer their incredible resources and dedication to the trade. I've learned so much from you, not just about geospatial analysis but also the value of geographic information, analysis, and generosity.

Some of this book is adapted from Qiusheng Wu's extensive, innovative, and useful geospatial resources, such as Geemap and Leafmap. Dr. Wu's vast geospatial knowledge, tutorials, videos, and courses are an incredible learning resource. Thank you, Qiusheng! I only hope this complements your work.

A special thank you to my family while I wrote this in the early AM or late PM hours. Your love and support have always inspired me. A special extra thank you to my spouse, who inspired me to become more serious about geospatial analysis. It's hard to start doing this when you have a Ph.D. in the house with a degree in ecological remote sensing, but you always encouraged me and, with some sighing, helped explain many things. Thank you, Emma!

you up to speed. Each chapter will offer resources and tutorials to help you get up to speed.

2.b. Chapters

The book contains the following chapters:

1. **Introduction.** This chapter.
2.  **FOSS.** This is an introduction to Free and open-source software, a little about my journey in this field, and my use of the tools in the book.
3.  **Google Earth Engine.** Google Earth Engine is an incredible resource for its large data catalog and cloud-based geospatial analysis. This chapter provides some quick tips and steps to get started.
4.  **Pros & Cons.** Outlines the pros and cons of free and open source vs. paid geospatial software.
5.  **Python.** How to use Python for geospatial data analysis and visualization with various libraries.
6.  **SQL.** The universal database Structured Query Language (SQL) with a focus on DuckDB.
7.  **QGIS.** An introduction to get started with this powerful desktop GIS.
8.  **AIML.** Artificial intelligence (AI) and machine learning (ML) are popular right now, right? Here's how to use these tools to complement, not replace, your work.
9.  **R.** A programming language universally adopted by academics, R is easy to use and start with and has many statistical computing, data visualization, and geospatial packages.
10.  **Field Surveys.** Your guide to open-source field data collection.
11.  **Drones.** Using drones for field survey and developing high resolution geospatial data. Chapter in draft, coming soon.
12.  **GRASS.** Using GRASS GIS in your workflows and data analysis. Chapter in draft, coming soon.
13.  **Visualization.** Effective principles for geospatial data visualization
14.  **Future.** Where your journey might lead with other established free and open-source geospatial software. We'll also explore the future of free and open-source geospatial software.

2.c. MyST

This book was made using [MyST](#), a free and open-source package that lets anyone build beautiful, publication-quality books and articles from computational content. Thank you, MyST for this wonderful resource!

3. BOOK CITATION

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